Name_______________________________Date_____________

Angle of Elevation & Depression Trig Worksheet

***Draw and label a picture for each problem**

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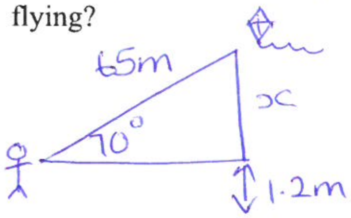
1. Brian's kite is flying above a field at the end of 65 m of string. If the angle of elevation to the kite measures 70°, and Brian is holding the kite 1.2 m off the ground. How high above the ground is the kite flying?



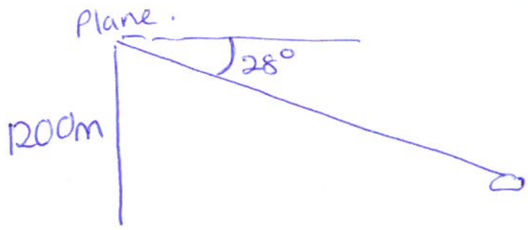
$\sin(70) = \frac{x}{65}$

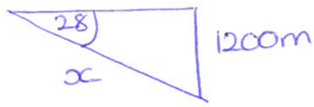
$x = 65 \times \sin(70)$

$x = 61.08m$

Total height $= 61.08 + 1.2$
$= 62.28m.$

2. From an airplane at an altitude (height) of 1200 m, the angle of depression to a rock on the ground measures 28°. Find the distance from the plane to the rock.



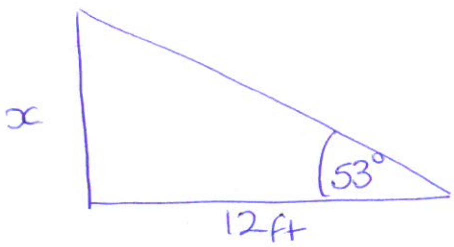


$\sin(28) = \frac{1200}{x}$

$x = \frac{1200}{\sin(28)}$

$x = 2556.07m$ (2dp).

3. From a point on the ground 12 ft from the base of a flagpole, the angle of elevation of the top of the pole measures 53°. How tall is the flagpole?

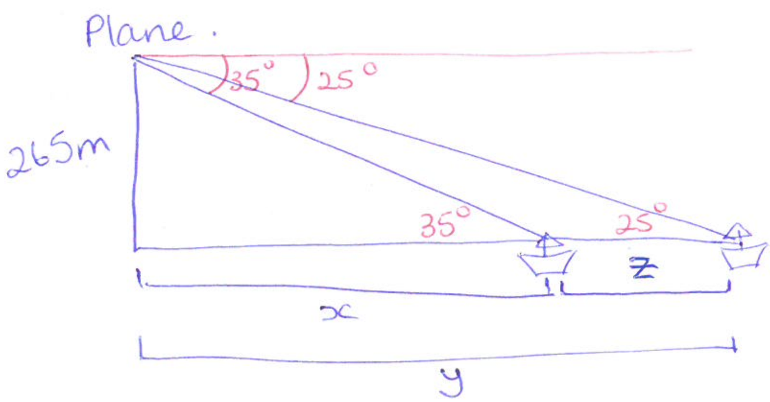


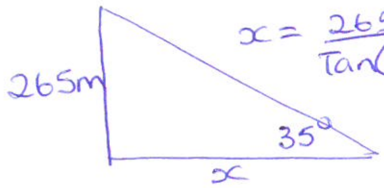
$\tan(53) = \frac{x}{12}$

$x = \tan(53) \times 12$

$= 15.92$ ft.

4. From a plane flying due east at 265 m above sea level, the angles of depression of two ships sailing due east measure 35° and 25 °. How far apart are the ships?

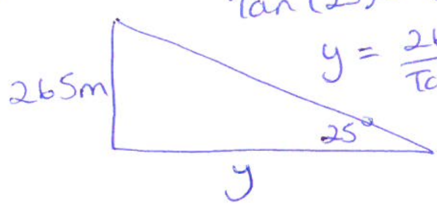




$\tan(35) = \frac{265}{x}$

$x = \frac{265}{\tan(35)}$

$= 378.46m$



$\tan(25) = \frac{265}{y}$

$y = \frac{265}{\tan(25)}$

$= 568.29m$

$z = y - x = 189.83m$

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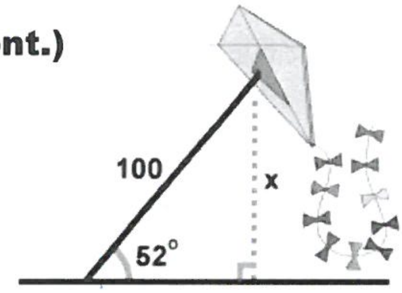
Angle of Elevation & Depression Worksheet (Cont.)

Find all values to the nearest tenth.

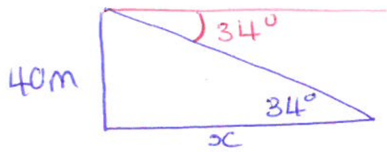
5. A man flies a kite with a 100 foot string. The angle of elevation of the string is 52°. How high off the ground is the kite?



$\sin(52) = \frac{x}{100}$

$x = 100 \times \sin(52) = 78.8\text{m}$

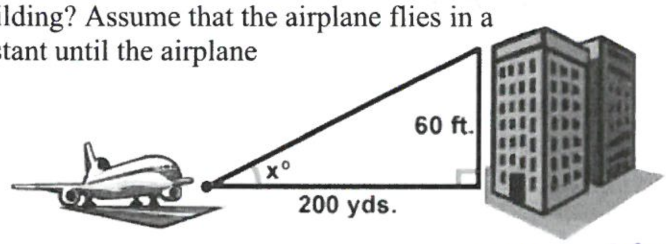
6. From the top of a vertical cliff 40 m high, the angle of depression of an object that is level with the base of the cliff is 34°. How far is the object from the base of the cliff?



$\tan(34) = \frac{40}{x}$

$x = \frac{40}{\tan(34)} = 59.3\text{m}.$

7. An airplane takes off 200 yards in front of a 60 foot building. At what angle of elevation must the plane take off in order to avoid crashing into the building? Assume that the airplane flies in a straight line and the angle of elevation remains constant until the airplane flies over the building.



$\tan(x) = \frac{20}{200}$

$\tan^{-1} = \frac{20}{200} = 5.7^\circ$

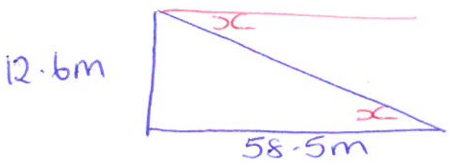
60ft = 20yds.

8. A 14 foot ladder is used to scale a 13 foot wall. At what angle of elevation must the ladder be situated in order to reach the top of the wall?

$\sin(x) = \frac{13}{14}$

$\sin^{-1} = \frac{13}{14} = 68.2^\circ$

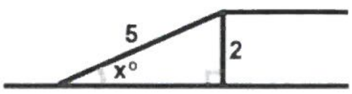
9. A person stands at the window of a building so that his eyes are 12.6 m above the level ground. An object is on the ground 58.5 m away from the building on a line directly beneath the person. Compute the angle of depression of the person's line of sight to the object on the ground.



$\tan(x) = \frac{12.6}{58.5}$

$\tan^{-1} = \frac{12.6}{58.5} = 12.2^\circ$

10. A ramp is needed to allow vehicles to climb a 2 foot wall. The angle of elevation in order for the vehicles to safely go up must be 30° or less, and the longest ramp available is 5 feet long. Can this ramp be used safely?



$\sin(x) = \frac{2}{5}$

$\sin^{-1} = \frac{2}{5} = 23.6^\circ$

$23.6^\circ < 30^\circ$

$\therefore$ Yes, the ramp can be used safely.